The digital world

Most of the devices we use for communicating and for storing information are digital. These include computers, cameras, and radios. Inside a digital device, information is stored and processed as numbers.

Digitizing information

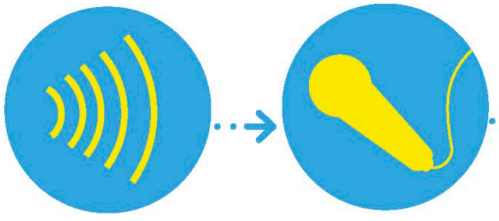
The information digital devices store and process includes text, images, sound, and video—and the software that makes the devices work. This information is represented by binary numbers, which consist of two digits: 0 and 1. Any number can be represented by a set of binary digits, or bits. Creating a representation of information like this is called digitization.

Why binary?
Inside digital devices, the binary digits, 0 and 1, typically exist as electric currents (off and on) or electric charges (present or absent). Embedded in all digital devices are computers, which store and process these numbers.

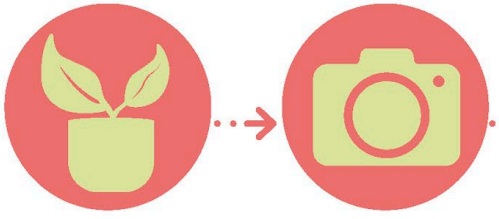
Digitizing touch
The touch screen of a smartphone (see pp.204–205) or tablet produces two binary numbers that represent the coordinates of the point on the screen you touch.



Digitizing sound
A circuit called an analogue-to-digital converter produces a stream of numbers that match the levels of voltage in an audio signal from a microphone (see pp.138–141) or musical instrument.



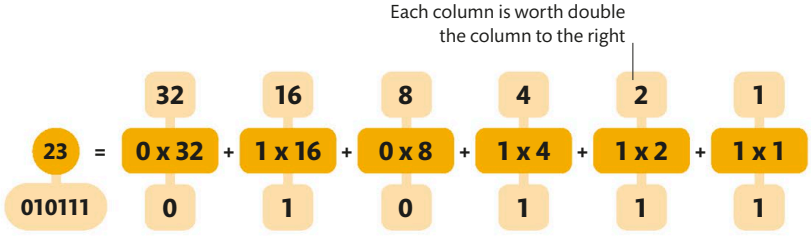
Digitizing images
A sensor inside a digital camera (see pp.152–153) produces numbers that correspond to the brightness of light at each picture element, or pixel, of an image.



Binary numbers

The binary number system is a place value system, just like the decimal system (0–9) that we use every day. But rather than 1s, 10s, 100s, 1,000s, and so on, the place values in the binary number system are 1s, 2s, 4s, 8s, and so on. Inside digital devices, electronic circuits produce electrical signals that represent binary digits, or bits. Most information is broken down into bytes: groups of eight bits.

Converting to binary
This example shows how the number we know in decimal as 23 can be expressed in binary.



THE **BINARY NUMBER SYSTEM** WAS DEVELOPED IN THE **17TH CENTURY**—LONG BEFORE IT WAS USED IN COMPUTING

